

2.1 Mathematical Background

Consider a two-layer neural network

$$h(x; \theta) = \frac{1}{\alpha(m)} \sum_{i=1}^m a_i \sigma(w_i^\top x), \quad (3)$$

where $x \in \mathbb{R}^d$, $w_i \in \mathbb{R}^d$ are the inner weights, $a_i \in \mathbb{R}$ are the outer weights, $\sigma(\cdot)$ is the activation function, and $\alpha(m)$ is a scaling factor defining the regime. At initialization, $W_{ij} \sim \mathcal{N}(0, 1)$, with $\mathbb{E}[a_i] = 0$ and $a_i \perp W_{ij}$, so that $\mathbb{E}[h(x; \theta)] = 0$ and $\text{Var}(w_i^\top x) = 1$. For n inputs $X \in \mathbb{R}^{n \times d}$ with $X \sim \mathcal{N}(0, I_d/d)$, (3) generalizes naturally to matrix form where $W \in \mathbb{R}^{m \times d}$. The three regimes differ only in $\alpha(m)$, and are designed to keep $\text{Var}(h(x))$ at a controlled scale as $m \rightarrow \infty$.

2.1.1 Mean Field (MF)

Setting $\alpha(m) = m$ gives $\text{Var}(h_{\text{MF}}(x)) = \frac{1}{m} \text{Var}(a_i)$ and $\text{Var}(\sigma(w_i^\top x)) = \frac{1}{m} \cdot 1 \cdot 1 = O(\frac{1}{m})$, so the variance of the network shrinks to zero as $m \rightarrow \infty$. Therefore by the law of large numbers, $\lim_{m \rightarrow \infty} h_{\text{MF}}(x) = \mathbb{E}[\Psi]$, where $\Psi(a, w): x \mapsto a\sigma(x^\top w)$ [1, 6].

2.1.2 Neural Tangent Kernel (NTK)

Setting $\alpha(m) = \sqrt{m}$ [2] keeps the variance stable:

$$\text{Var}(h_{\text{NTK}}(x)) = m \cdot \frac{1}{m} \text{Var}(a_i) \text{Var}(\sigma(w_i^\top x)) = O(1),$$

so unlike MF, the output variance doesn't vanish with width. Both inner and outer weights are trained.

2.1.3 Random Features (RF)

The RF regime [8] uses the same scaling $\alpha(m) = \sqrt{m}$, and therefore has the same $O(1)$ output variance as NTK. The key distinction is that only the outer weights a_i are trained; the inner weights w_i are fixed at initialization. This reduces (3) to a linear model with random features $\Phi_i(x) = \sigma(w_i^\top x)$.

2.2 Experiment 1: Performance Comparison Across Regimes

2.2.1 Objective

This experiment compares training and generalization performance across all three regimes (MF, NTK, RF) to understand which regime converges fastest, which activation function performs best, and how performance scales with network width m . We expect Mean Field to perform the best as its learning rate scales with width and Random Features to perform the worst as it learns less features.

2.2.2 Experimental Setup

The network (3) was evaluated on three datasets: a synthetic dataset, MNIST [4], and CIFAR-10 [3].

Synthetic Dataset Labels were generated as $y_j = h^*(x_j)$, where h^* is a ground-truth network of the same form as (3) with fixed parameters $\{a_i^*, w_i^*\}$ and $\sigma^* = \text{ReLU}$. Inputs were drawn as $x_j \sim \mathcal{N}(0, I_d/d)$, ensuring $\mathbb{E}[\|x_j\|_2^2] = 1$ across all dimensions.

CIFAR-10 and MNIST Pre-processing Both datasets were reduced to binary classification (classes 0 & 1) and images were flattened to $d = 784$ (MNIST) and $d = 3072$ (CIFAR-10). Each sample was then standardized as $x_j \leftarrow \frac{x_j - \bar{\mu}}{\bar{\sigma}\sqrt{d}}$, so that $\mathbb{E}[\|x_j\|_2^2] \approx 1$, matching the $\mathcal{N}(0, I_d/d)$ assumption of (3).

Evaluation Framework All regimes were trained across $\sigma \in \{\text{ReLU}, \text{erf}, \text{tanh}\}$, widths $m \in \{100, 200, 400, 800\}$, and $T = 1000$ iterations. Synthetic experiments used $d = 10$, $n = 3000$, $n_{\text{test}} = 500$, $m^* = 20$; real datasets used $n = 4000$, $n_{\text{test}} = 1000$. Learning rates $\beta_{\text{MF}} \in \{0.03, 0.1, 0.5\}$ and $\beta_{\text{NTK}}, \beta_{\text{RF}} \in \{0.1, 0.5, 1.0\}$ were tested. According to Section 2.3, $\beta_{\text{MF}} = 0.1, \beta_{\text{NTK}} = \beta_{\text{RF}} = 1.0$ are the optimal parameters for each regime's learning rate hence why they are included. The other choices of β are tested for comparison. All experiments minimized the mean squared error (MSE) loss and were run with seed = 42.